MORPHISMS BETWEEN REPRESENTABLE FIBRED CATEGORIES MODULO 2-ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 8698-8716, following the statement at 8678-8696.

Editorial note. The proof is a complete chain of three instantiations; the transport of Mor/2-iso classes through the representability equivalences is standard 2-categorical bookkeeping recorded in the card's obligations.

Editorial note. The cited lemma-2-category-fibred-setoids is a standing input; its own proof was judged uncardable separately and is not re-verified in this span.

Allowed standard prerequisites (editorial).

- Fibred category associated to a presheaf. example-fibred-category-from-functor-of-points, cited at proof lines 8699-8702. Supplies the identification of the representable fibred categories with the categories fibred in sets associated to h_X and h_Y.
- 2-category of fibred setoids versus presheaves. lemma-2-category-fibred-setoids (its final assertion), cited at proof lines 8702-8711. Supplies the bijection Mor_Cat/C(X,Y)/2-iso = Mor_PSh(F_X, F_Y) for categories fibred in setoids, applied to the representable (hence setoid) fibred categories.
- Yoneda lemma. lemma-yoneda, cited at proof lines 8712-8715. Supplies Mor_PSh(h_X, h_Y) = Mor_C(X, Y).

Source reference. Original reference 0044: sources/stacks-categories/pinned-original-source.tex, source lines 8326–8326. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04SC: sources/stacks-categories/pinned-original-source.tex, source lines 8489–8489. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 001P: sources/stacks-categories/pinned-original-source.tex, source lines 758–758. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:categories.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{C}$ be a category. Let $\mathcal{X}$, $\mathcal{Y}$ be categories fibred in groupoids over $\mathcal{C}$. Assume that $\mathcal{X}$, $\mathcal{Y}$ are representable by objects X, Y of $\mathcal{C}$. Then*

$$\mathrm{Mor}_{Cat/\mathcal{C}}(\mathcal{X}, \mathcal{Y}) \Big/ 2\text{-}\textit{isomorphism} = \mathrm{Mor}_{\mathcal{C}}(X, Y)$$

More precisely, given $\phi : X \to Y$ there exists a 1-morphism $f : \mathcal{X} \to \mathcal{Y}$ which induces ϕ on isomorphism classes of objects and which is unique up to unique 2-isomorphism.

Proof. By Example 0044 we have $\mathcal{C}/X = \mathcal{S}_{h_X}$ and $\mathcal{C}/Y = \mathcal{S}_{h_Y}$. By Lemma 04SC we have

$$\mathrm{Mor}_{Cat/\mathcal{C}}(\mathcal{X}, \mathcal{Y}) \Big/ 2\text{-isomorphism} = \mathrm{Mor}_{PSh(\mathcal{C})}(h_X, h_Y)$$

By the Yoneda Lemma 001P we have $\mathrm{Mor}_{PSh(\mathcal{C})}(h_X, h_Y) = \mathrm{Mor}_{\mathcal{C}}(X, Y)$. □